03 November 2025 08:34 AM

1.Solution:

The LPP can now be written as:

$$\text{Maximize } Z = -20x + 50y$$

subject to:

$$x \geq 150,$$

$$y \geq 100,$$

$$x \leq 250,$$

$$y \leq 200,$$

$$x + y \geq 250,$$

$$x \geq 0,\ y \geq 0.$$

2.Solution

The LPP is as follows:

$$\text{Minimize } Z = 6x + 3y$$

Subject to:

$$3x + 5y \geq 50$$

$$4x + 3y \geq 60$$

$$x \geq 0,\ y \geq 0$$

3. Solution:

The LPP is :

$$Min\ Z = 40x_1 + 80x_2$$

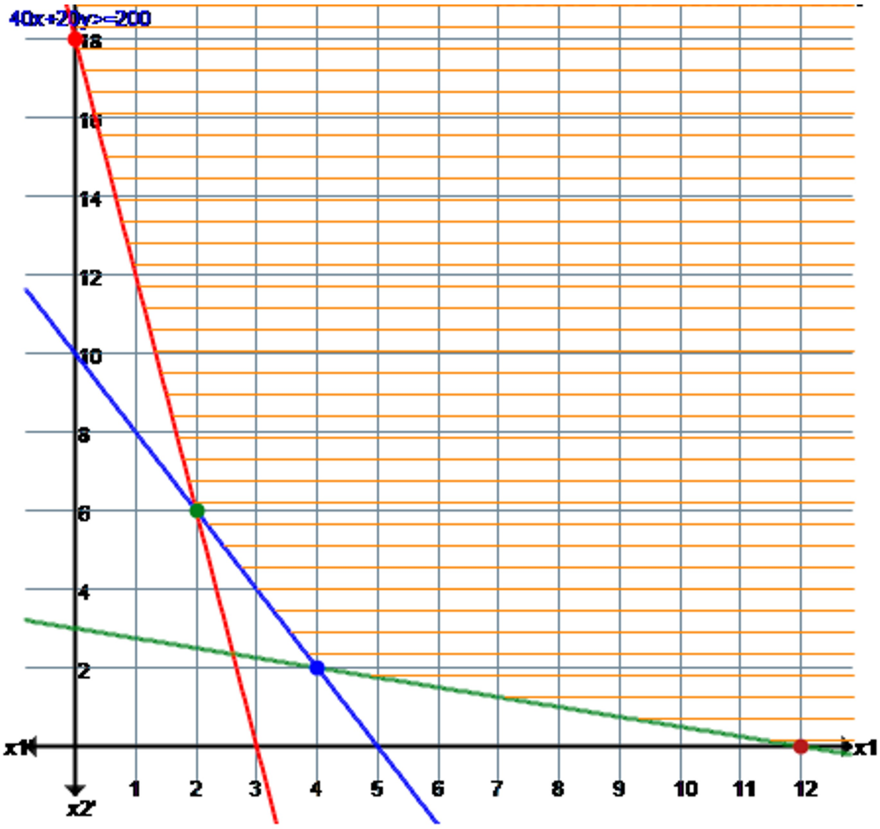
Subject to

$$72x_1 + 12x_2 \geq 216$$

$$6x_1 + 24x_2 \geq 72$$

$$40x_1 + 20x_2 \geq 200 \quad and \quad x_1 \geq 0, x_2 \geq 0.$$

Graphical solution is



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1,x_2)	Lines through Extreme Point	Objective function value $Z = 40x_1 + 80x_2$
$A(0, 18)$	$1 \to 72x_1 + 12x_2 \geq 216$ $4 \to x_1 \geq 0$	$40(0) + 80(18) = 1440$
$B(2, 6)$	$1 \to 72x_1 + 12x_2 \geq 216$ $3 \to 40x_1 + 20x_2 \geq 200$	$40(2) + 80(6) = 560$
$C(4, 2)$	$2 \to 6x_1 + 24x_2 \geq 72$ $3 \to 40x_1 + 20x_2 \geq 200$	$40(4) + 80(2) = 320$
$D(12, 0)$	$2 \to 6x_1 + 24x_2 \geq 72$ $5 \to x_2 \geq 0$	$40(12) + 80(0) = 480$

The miniimum value of the objective function $Z = 320$ occurs at the extreme point $(4, 2)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 4, x_2 = 2$ and min $Z = 320$.

4.

The Linear Programming Problem is:

$$\text{Maximize } Z = 160x_1 + 240x_2$$

Subject to:

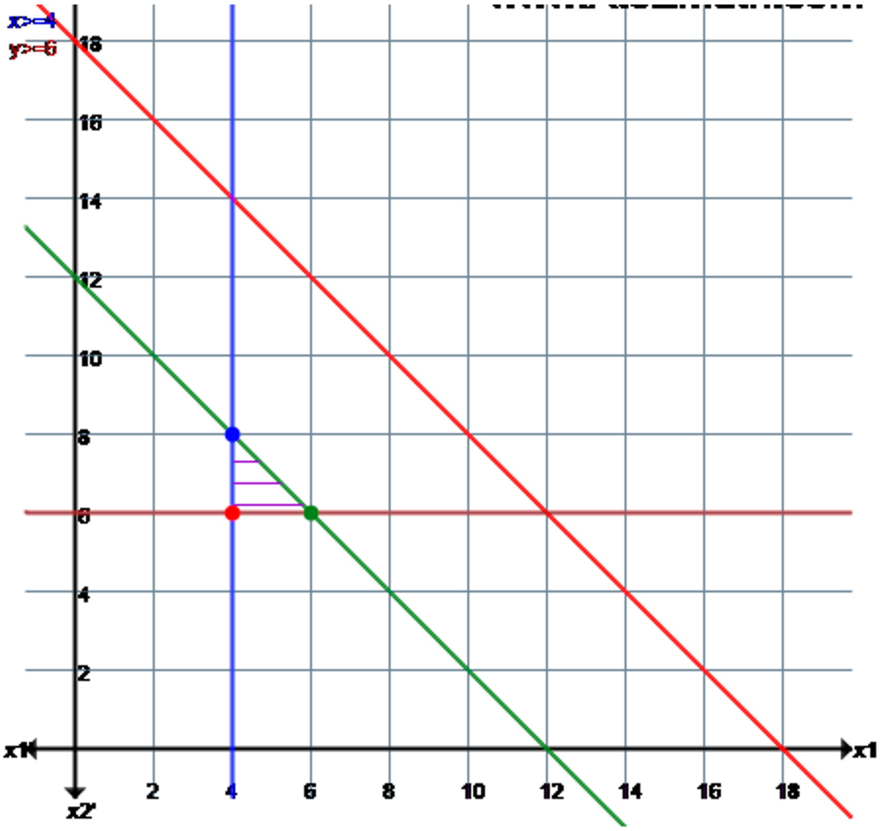
$$x_1 + x_2 \leq 18$$

$$x_1 + x_2 \leq 12$$

$$x_1 \geq 4$$

$$x_2 \geq 6$$

$$x_1 \geq 0,\, x_2 \geq 0$$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = 160x_1 + 240x_2$
$A(4, 6)$	$3 \to x_1 \geq 4$ $4 \to x_2 \geq 6$	$160(4) + 240(6) = 2080$
$B(6, 6)$	$2 \to x_1 + x_2 \leq 12$ $4 \to x_2 \geq 6$	$160(6) + 240(6) = 2400$
$C(4, 8)$	$2 \to x_1 + x_2 \leq 12$ $3 \to x_1 \geq 4$	$160(4) + 240(8) = 2560$

The maximum value of the objective function $Z = 2560$ occurs at the extreme point $(4, 8)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 4, x_2 = 8$ and max $Z = 2560$.